Candidate Seat no............

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (EC) Examination

March 2018

EC301.01/301 Electromagnetic Theory

Date: 27.03.2018, Tuesday **Time: 1:30 p.m. to 4.30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Use of non-programmable scientific calculator is allowed.

SECTION-I

Q-1 Answer Following Questions **[05]**

(a) Define Scalar and Vector. Give one example of each. [02]

(b) Explain Cartesian Coordinate System. [03]

Q-2 Attempt Any Three **[15]**

(a) Explain Current Continuity equation. [05]

(b) Illustrating figure(s), perform step by step analysis for conversion from Spherical coordinate system to Cartesian Coordinate system [05]

(c) Describe Poisson and Laplace Equations [05]

(d) Explain method of direct integration for parallel plate capacitor. [05]

(e) Given the points A($x=2, y=3, z=-1$) and B($r = 4$, $\theta = 25^0$, $\phi=120^0$), calculate (i)Spherical coordinates of **A**; (ii) the Cartesian coordinates of **B** (iii) The distance from **A** to **B**. [05]

Q-3 Attempt Any Three **[15]**

(a) Explain Poynting's Theorem. [05]

(b) State and Prove Uniqueness Theorem. [05]

(c) Describe Wave Propagation in Vacuum. [05]

(d) Describe Gradient, Divergence and Curl with example. [05]

(e) An aluminum conductor is 1000 feet long and has a circular cross section with a diameter of 0.8in. If there is a dc voltage of 1.2 V between the ends, find: (i)the current density; (ii) the current; (iii) the power dissipated. [05]

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SECTION – II

Q-4 For vector field A, prove: $\nabla \times \nabla \times A = \nabla(\nabla . A) - \nabla^2 A$. **[07]**

Q-5 Attempt Following Questions **[14]**

(a) An electric field is given as $\vec{E} = 6y^2 z\, a_x + 12\, x\, y\, z\, a_y + 6\, x\, y^2\, a_z\ V/m$. An incremental path is represented by $\overrightarrow{\Delta L} = -3\, a_x + 5\, a_y - 2\, a_z\ \mu m$. Find the work done in moving a 2 μC charge along this path if the location of the path is at (i) $P_A(0, 2, 5)$ (ii) $P_B(1, 1, 1)$ (iii) $P_C(-0.7, -2, -0.3)$. [07]

(b) Find $\nabla \times \vec{G}$, [07]

(i) In Cartesian coordinates at $P_A(3, 2, 1)$ if $\vec{G} = xyz\ (\overrightarrow{a_x} + \overrightarrow{a_y})$,

(ii) In cylindrical coordinates at $P_B(2, 30^\circ, 3)$ if $\vec{G} = \rho \emptyset z\ \overrightarrow{a_z}$,

(iii)In spherical coordinates at $P_C(4, 30^\circ, 45^\circ)$ if $\vec{G} = \sin\theta\ \ (\overrightarrow{a_r} + \overrightarrow{a_\theta} + \overrightarrow{a_\emptyset})$.

OR

Q-5 Attempt Following Questions **[14]**

(a) Find a numerical value for the divergence of $\vec{D}$ at the point indicated if [07]

(i)$\vec{D} = 20x\, y^2\, (z+1)\, \overrightarrow{a_x} + \ 20\, x^2 y\, (z+1)\, \overrightarrow{a_y} + 10\, x^2 y^2 \overrightarrow{a_z}\ C/m^2$ at $P_A(0.3, 0.4, 0.5)$,

(ii) $\vec{D} = 4\,\rho\, z \sin\emptyset\ \overrightarrow{a_\rho} + \ 2\,\rho\, z \cos\emptyset\ \overrightarrow{a_\emptyset} + 2\,\rho^2 sin\emptyset\ \overrightarrow{a_z}\ C/m^2$ at $P_B(1, \pi/2, 2)$,

(iii)$\vec{D} = \sin\theta\ \cos\emptyset\ \ \overrightarrow{a_r} + \ \cos\theta \cos\emptyset\ \overrightarrow{a_\theta} - sin\emptyset\ \overrightarrow{a_\emptyset}\ C/m^2$ at $P_C(2, \frac{\pi}{3}, \frac{\pi}{6})$.

(b) State the divergence theorem. Start from the gauss's law and prove the divergence theorem. [07]

Q-6 Attempt Following Questions **[14]**

(a) Elaborate the magnetic field intensity for the infinite length current filament. [07]

(b) Derive the electric field intensity ($\vec{E}$) for infinite sheet charge. [07]

OR

Q-6 Attempt Following Questions **[14]**

(a) Define gradient and list out gradient of scalar entity in Cartesian, coaxial and spherical coordinate systems. [07]

(b) Enlist time varying Maxwell's equations in integral form and point form. [07]

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